

SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Compact VLMs that achieve competitive performance with dramatically lower memory footprints through strategic architectural optimizations.

**Sub-1GB Inference | 256M / 500M / 2.2B
Parameters**

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs (e.g., Idefics-80B) require tens of gigabytes of GPU memory, making mobile/edge deployment infeasible.

Existing efficient open-source models still consume 10–30 GB RAM, far exceeding smartphone and IoT device limits.

The gap between research-scale models and on-device inference creates a bottleneck for real-world multimodal applications.

Key challenge: achieve competitive vision-language understanding without sacrificing the memory and latency constraints of edge hardware.

Idefics-80B

80B

>> 30 GB

Molmo-A1B-7B

7B

27.7 GB

InternVL2-2B

2B

~10 GB

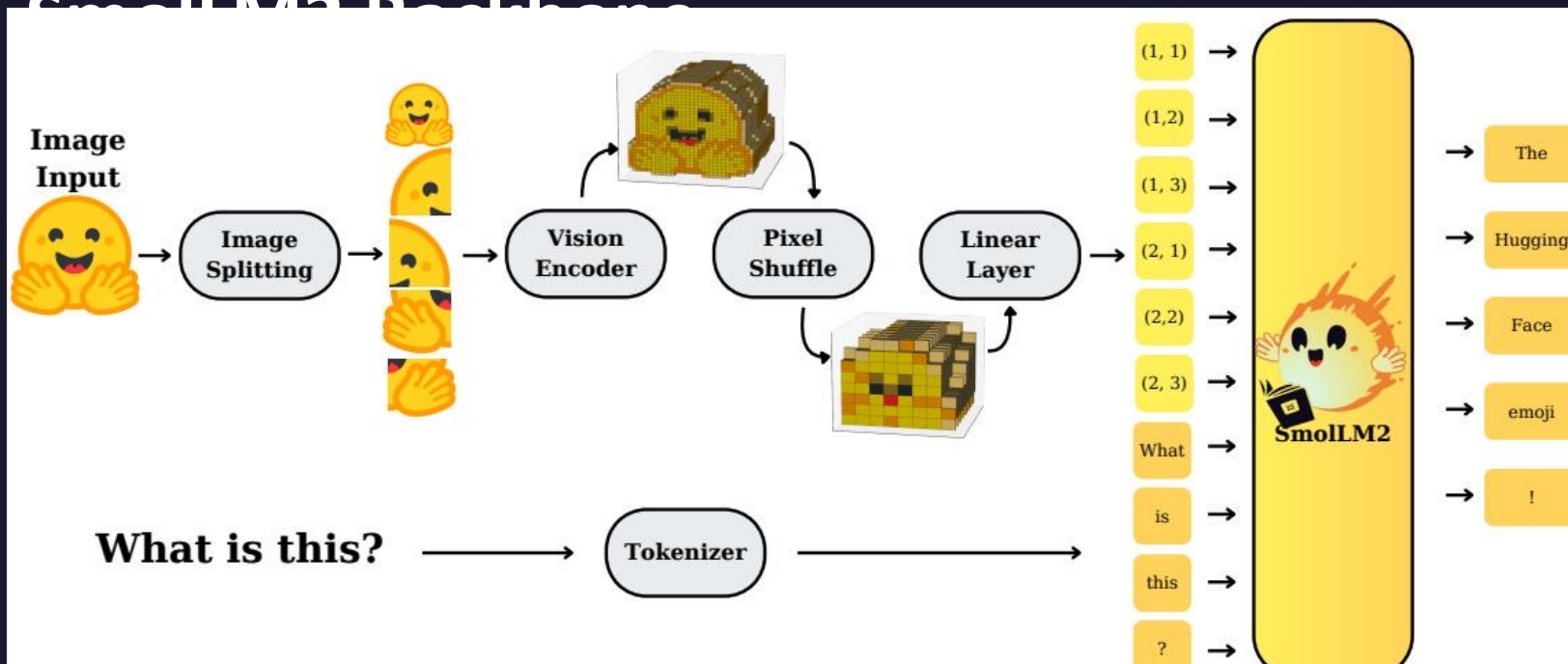
SmoVLM-256M

256M

0.8 GB

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and

SmolLM2 Backbone



1. Image Splitting

Input image is divided into sub-images for local processing.

2. Vision Encoder

SigLIP encoder extracts patch-level visual features.

3. Pixel Shuffle

Spatial features rearranged into channels (r^2 token reduction).

4. Linear Projection

Visual tokens projected into SmolLM2 input embedding space.

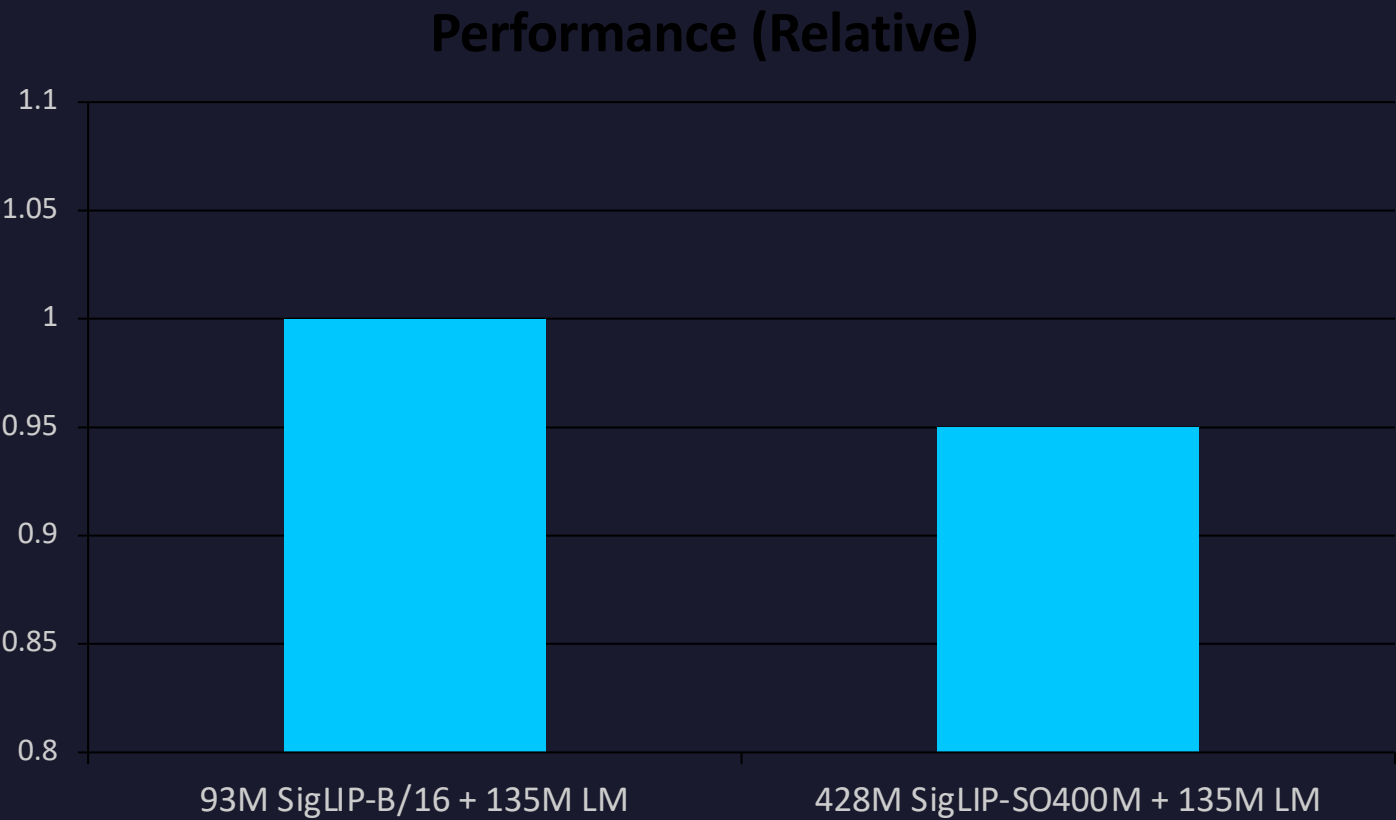
5. SmolLM2 Backbone

Compact language model fuses vision and text for generation.

Smaller Vision Encoders Outperform Larger Ones in Compact

VLMs

Finding 1: Balanced encoder-LM parameter allocation matters. A 93M SigLIP-B/16 paired with a 135M LM outperforms a 428M SigLIP-SO400M with the same LM.



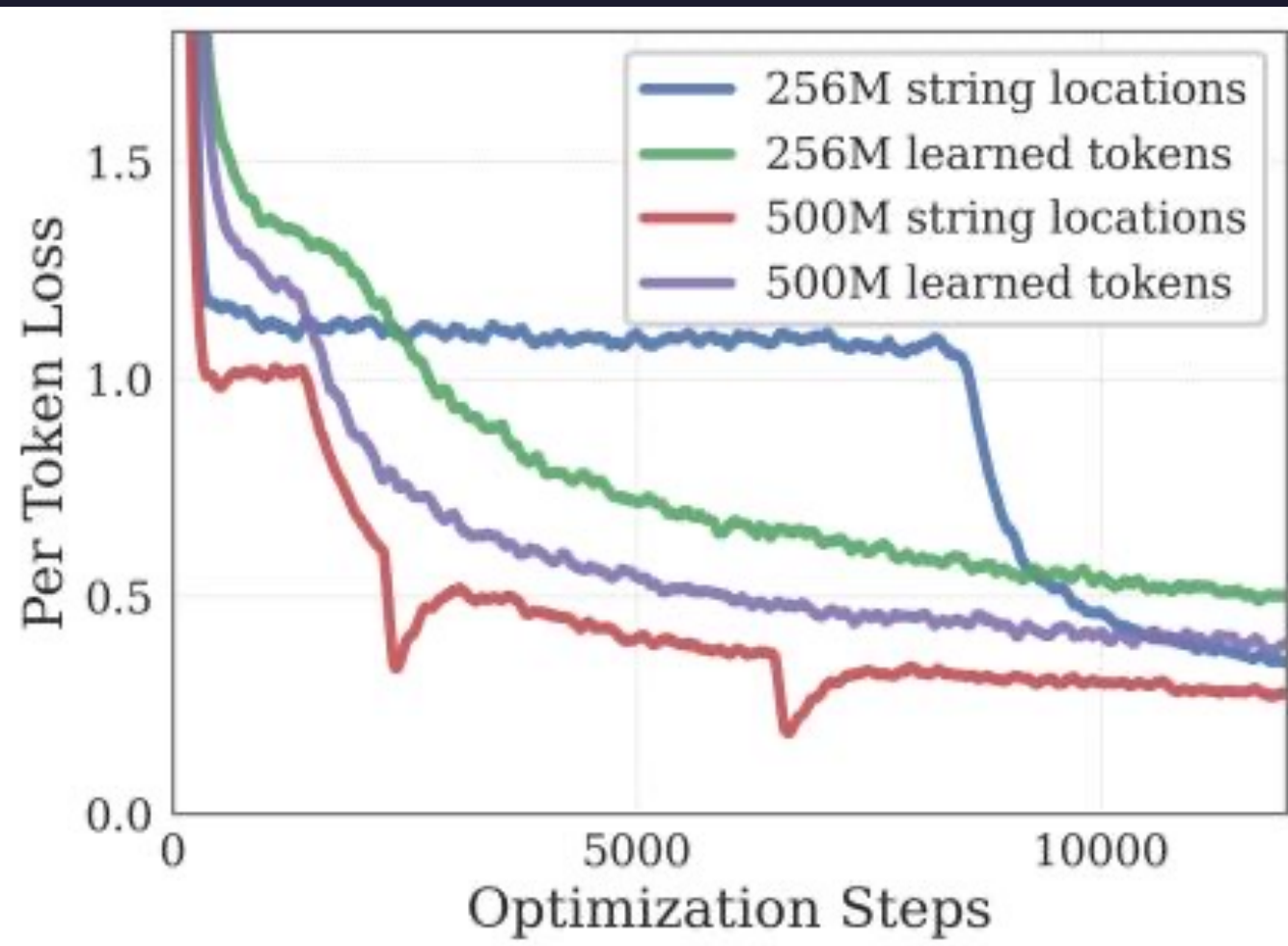
The larger encoder only justifies its 10% parameter overhead at 1.7B LM scale.

Encoder Comparison

Encoder	Params	Best For
SigLIP-B/16	93M	Compact VLMs (<1B LM)
SigLIP-SO400M	428M	Large VLMs (>1.7B LM)

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Pixel shuffle rearranges spatial features into channels, reducing visual tokens by r^2 . Small VLMs benefit from $r=4$ (16× reduction); larger models prefer $r=2$ (4×).



Token Reduction Factor

$r = 2$

4× reduction

Larger models (2.2B)

$r = 4$

16× reduction

Small models (256M, 500M)

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Image 2 performance improves significantly when extending context from 2K to 16K tokens. Achieved by increasing the RoPE base from 10K to 273K and fine-tuning on mixed long/short-context data.

2K Tokens

Baseline context length

Limited image resolution

8K Tokens

SmolVLM-256M / 500M

Higher resolution, multi-image

16K Tokens

SmolVLM-2.2B

Full high-resolution images + video

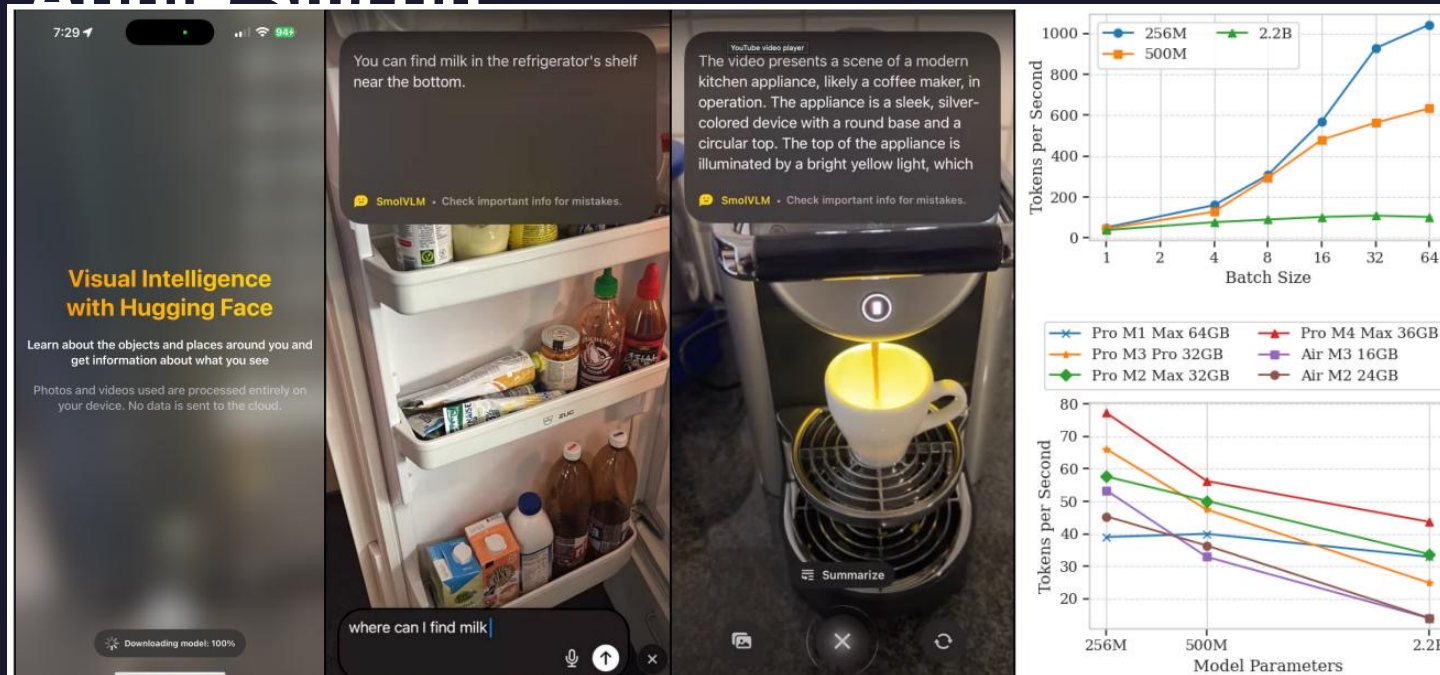
Technical implementation: RoPE base increased from 10,000 → 273,000. Fine-tuning on mixed long/short-context data ensures stability across sequence lengths. The 2.2B model uses 16K context; smaller variants use 8K.

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
Average	All Benchmarks	44.0%	51.0%	59.8%	—

RAM Usage: SmolVLM-256M = 0.8 GB | SmolVLM-500M = 1.2 GB | SmolVLM-2.2B = 4.9 GB | Molmo-A1B-7B = 27.7 GB

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



Throughput Benchmarks

~80 tok/s

Pro M1 Max 64GB

Batch = 1

~1000 tok/s

High-end devices

Batch = 64

~55 tok/s

MacBook Air M3 16GB

Batch = 1

Throughput scales roughly linearly with batch size for the 256M model.

Key Takeaways: Design Principles for Efficient Multimodal Models

1. Balance Encoder-LM Compute

Compact VLMs benefit from smaller vision encoders. A 93M SigLIP-B/16 outperforms a 428M SigLIP-SO400M when paired with small LMs.

2. Extend Context Length

Increasing RoPE base from 10K to 273K and fine-tuning on mixed data unlocks 8K–16K context, enabling higher-resolution inputs.

3. Use Aggressive Token Compression

Pixel shuffle at $r=4$ yields 16× token reduction for small models, easing attention overhead and improving long-context modeling.

4. Sub-1GB Inference Is Achievable

SmolVLM-256M runs at 0.8 GB RAM and surpasses the 300× larger LLaVA-72B — efficiency beats raw scale.

5. Fully Open-Source Release

All weights, datasets, code, and the HuggingSnap iOS app are open-sourced to accelerate on-device AI research.

Implication: Architectural efficiency, not just parameter count, is the key to practical on-device multimodal AI.